Sammy Al Hashemi

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(647)-463-2638

Education

- University of Toronto
  - Bachelor of Applied Science and Engineering
- Department of Engineering Science
- Major in physics

cGPA: 3.56

Software Skills

- Python
- Javascript/nodeJS
- Angular
- ReactJS

Hardware Skills

- Verilog

Courses Taken

- Algorithms/Foundations of Computing
- Introduction to Databases
- Data Structures & Algorithms
- Machine Learning and Data Mining
- Natural Language Processing
- Electronic Circuits/Digital Electronics

Website

https://website2-e5cf5.firebaseapp.com

Work Experience

Device Characterization and Product Engineering Intern

Microsemi Corp.

July 2017 - August 2018

- Characterized the Mathblock component of Microsemi's G5 FPGA product as well as performance-related AC tests on other combinatorial and sequential elements found in the FPGAs that went to market.
  - Example components include look-up tables, carry chains, digital I/O's, characterized mainly for max frequency.
  - Tested chains on different corners of the FPGA fabric.
- Inside Libero, I created Verilog test structures, making use of JTAG to read input from test programs created using a company code.
- Wrote sections of the characterization report found in the official product manual used to facilate customer knowledge in our device.

Projects

Thesis

- Using IBM Q's python SDK, `qiskit`, I am programming scalable versions of Grover's algorithm for my thesis under Proffessor Glen Gulak.
- I will be presenting my results in April 2019.

Data Charter

- Built a charting application that reads data that is interfaced from a *ni daqmx* board.
- The actual chart was built using *pyqt5* and is able to display any number of channels vs. time/other channel.
- In production at UofT's Advanced Physics Lab.

Personal Website

- Single-page application written in an Angular6 frontend with Typescript
- Uses Firebase's Cloud Storage to store and asynchronously fetch larger files such as pictures.
- Uses complex angular routing features such as lazy loading
- RxJS usage for asynchronous data handling (migrated from RxJS5 to RxJS6)

DashCam Speed Visualizer

- Comma.ai's dashcam product has the capability to sample data every second, of which includes speed, coordinates, and distance since the trip started.
- Using 527 trips, stored as `JSON`, I created a performant webapp that simultaneously displayed all 527 and their speed distribution on an interactive map. See https://dashcamroutes.herokuapp.com/
- I stored the trips in a Mongo Cloud Atlas database in three collections, all dependent on the zoom of the map.
  - Python was used to seperate these trips and uploaded them to the Cloud.
  - Client side caching was implemented so that the data did not need to be downloaded multiple times.
- An `Angular2+` frontend was implemented with an `ExpressJS` backend with `Mongoose` acting as the interface between Mongo and the app.